

Detection, and Avoidance

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Abstract

Leakage in data mining is the unintentional introduction of information about the target variable that should not legitimately be available at test/prediction time. This paper formalizes the taxonomy of leakage—distinguishing target leakage and train-test contamination—and introduces practical detection and avoidance protocols. It demonstrates how subtle pipeline leaks produce illusory high accuracy in offline evaluations that completely collapse in production deployment.

Role in Our VLM-DB Study

Cited alongside Grimm et al. in Section 1 and Section 3 to establish rigorous train/test separation and prove that multi-vector retrieval pipelines must strictly enforce learn-predict independence during evidence index construction.